UDAY GOEL

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SUMMARY

Physics & Quantum Computing student at the University of Pittsburgh with research experience in machine learning, large language models, and statistical modeling. Skilled in Python, R, and MATLAB, with hands-on work in time-series analysis and quantum computing.

EDUCATION

University of Pittsburgh **Expected Graduation: May 2030**
BS in Physics & Quantum Computing | Expected GPA: 3.6
Relevant Coursework: Calculus 3, Physics 1, Programming, Discrete Mathematics

Rock Ridge High School, Ashburn VA **August 2022 – June 2026**
Advanced Studies Diploma | GPA: 3.82
Relevant Coursework: AP Calculus AB/BC, AP Computer Science A, AP/DE Chemistry

RELEVANT EXPERIENCE

University of Michigan - Department of Statistics (LSA) **June 2025 – August 2025**
Research Assistant
• Conducted faculty-mentored research under Dr. Edward Ionides (Associate Chair, Statistics) on time-series modeling of cryptocurrency market volatility.
• Analyzed a 10-year Bitcoin dataset of 3,650 data points for volatility and stationarity using six time-series models (ARIMA, ARMA, ARCH, GARCH, EGARCH, GJR-GARCH) in R and MATLAB.
• Built and compared multivariable statistical models to identify key volatility patterns and presented results in a final research presentation.

Qwary - Capstone Research **May 2026 – June 2026**
LLM Efficiency Researcher
• Completed 50+ hours of independent research on Parameter-Efficient Fine-Tuning (PEFT) methods for large language models.
• Evaluated LoRA and QLoRA for hardware-efficient LLM deployment, comparing trade-offs in memory use and model performance.
• Synthesized academic literature and presented findings at the annual Capstone Gala & Showcase.

Virginia Polytechnic Institute, College of Science **July 2025**
Data Science Summer Camp
• Selected for a competitive summer data science program; cleaned and analyzed public datasets in R.
• Built regression models examining preferences for natural amenities across U.S. counties.
• Collaborated on a team research project and presented results to an evaluation committee.

Sharada University – Anatomy Education Research **March 2025**
Research Assistant (Data & Survey Analysis)
• Analyzed survey data comparing medical students' preferences for cadaveric versus virtual anatomy learning methods.
• Assessed student confidence and learning outcomes, supporting statistical analysis, interpretation, and presentation of findings.

LEADERSHIP EXPERIENCE

Rock Ridge High School **August 2022 – February 2026**
Captain (Senior Year), Congressional Debate Team Member
• Led the team as Captain, ran strategy sessions, mentored novice members, and drove recruitment.
• Qualified for the 2025 Metro Finals and accumulated 240+ career debate points.
• Researched and wrote legislative proposals on domestic, economic, and international policy issues.

Boy Scouts of America, Troop 1666 **August 2015 – April 2026**
Eagle Scout and Senior Patrol Leader
• Earned Eagle Scout rank; planned and led a community service Eagle Project for Crossroads United Methodist Church.
• Served in leadership roles including Senior Patrol Leader, Quartermaster, and Webmaster.

SKILLS & AWARDS

Technical Skills: Python, R, JavaScript, MATLAB, RStudio, Google Collab, Google Analytics, Claude Code, Codex,
Certifications: IBM Data Science; Google Data Analytics, Java for Beginners
Honors & Awards: R.I.S.E. Award; Science National Honor Society